Hall Ticket Number:

MR-21

Code: B413CS4

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Regular Examinations, Aug/Sep-2023

Java Programming

(Common to CSE & IT)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Compare Procedural oriented languages and OOP?	2	2	1	1,2
1. b)	What is an Object? How to allocate memory for objects?	2	1	1	1,2
1. c)	What happens when there is no suitable try block to handle exception?	2	1	2	1,2
1. d)	Define interface?	2	1	2	1,2
1. e)	What is light weight process? Discuss.	2	1	4	1,2
1. f)	Draw the thread life cycle.	2	1	4	1,2
1. g)	Explain Scanner class with an example.	2	2	3	1,2
1. h)	Define Java collection Framework.	2	1	3	1,2
1. i)	What is an Event? When will it occur?	2	1	5	1,2
1. j)	Explain about limitations of AWT?	2	2	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the advantages and disadvantages of Object-Oriented Programming.	5	2	1	1,2
2. b)	Differentiate between method overloading and method overriding with examples	5	4	1	1,2,3
	OR				
2. c)	List and explain java buzzwords.	5	2	1	1,2
2. d)	What is inheritance? Explain making use of inheritance in java with examples.	5	3	1	1,2,3
3. a)	Write a java program for user defined exception.	4	3	2	1,2,3
3. b)	What is package? Create a Java package and explain it.	6	1	2	1,2,3
	OR				
3. c)	What is interface? How to create it and apply it? Explain with example.	5	3	2	1,2,3
3. d)	What are the various types of exceptions available in Java? Also discuss on how they are handled?	5	1	2	1,2
4. a)	Explain with example, suspending and resuming threads in Java.	5	2	4	1,2,3
4. b)	Differentiate between FileReader and BufferedReader.	5	4	3	1,2
	OR				
4. c)	Explain the synchronization of multiple threads in Java with an example.	4	3	4	1,2,3
4. d)	Illustrate java.io package in detail.	6	2	3	1,2
5. a)	What is Java Collections Framework? List out some benefits of Collections framework and explain.	5	1	3	1,2
5. b)	What is an Iterator? Demonstrate how to access an Iterator using an example.	5	3	3	1,2,3
	OR				
5. c)	Compare and Contrast between ArrayList and Vector?	5	4	3	1,2
5. d)	What are the basic interfaces of Java Collections Framework?	5	1	3	1,2
6. a)	What is applet? Discuss about applet structure.	5	1	5	1,2
6. b)	Explain about components in swings.	5	2	5	1,2
	OR				
6. c)	What is AWT? Distinguish between AWT and swings	5	4	5	1,2
6. d)	Analyze the role of event listeners in event handling? List the Java event listeners	5	4	5	1,2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome